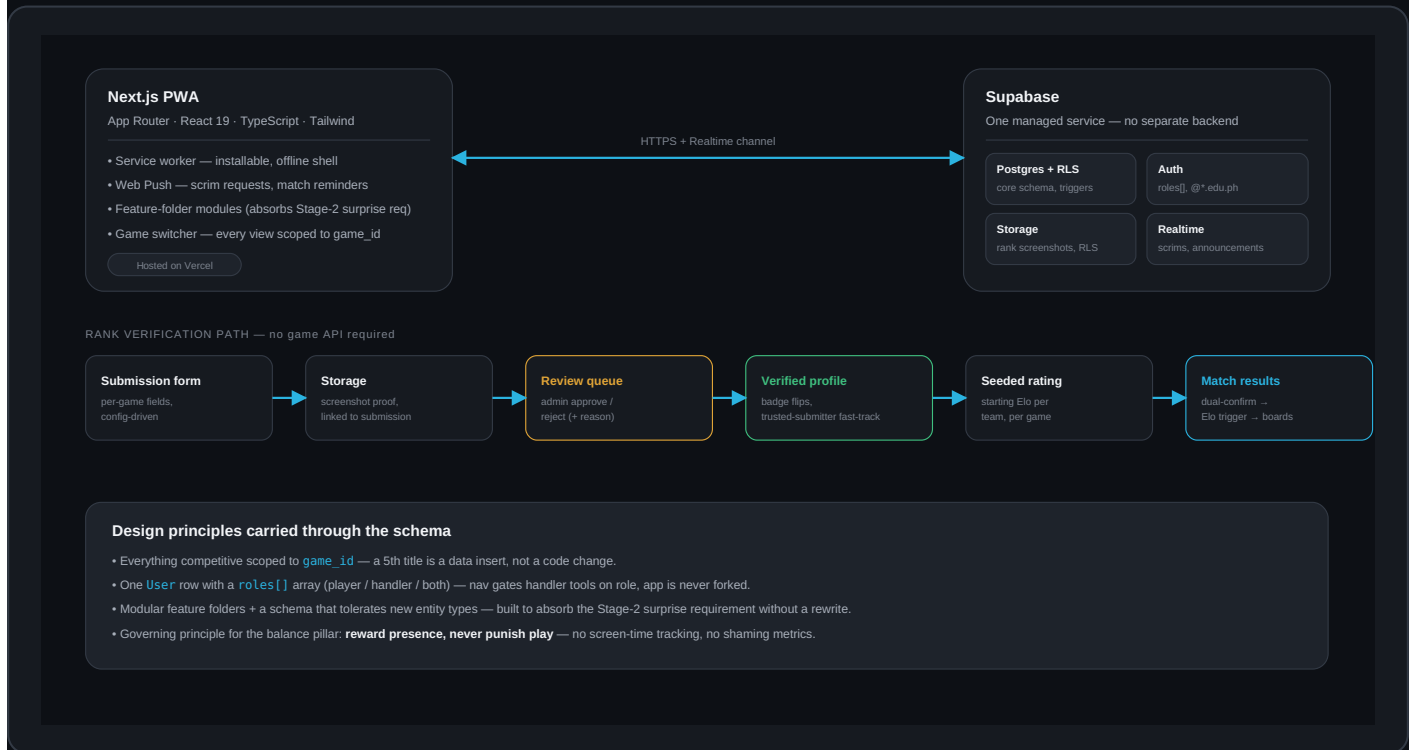


Solution Architecture

The scrim network for Philippine collegiate esports — a Next.js PWA on Vercel, backed end-to-end by Supabase.



How it fits together

Everything competitive is scoped to a `game_id` — scrim, teams, ranks, leaderboards, events, and announcements all carry one, and each game is a config row (rank tiers + per-game form fields), so a fifth title is a data insert, not a code change. Accounts are a single row carrying a `roles[]` array (player, handler, or both); navigation gates handler tools on the role rather than forking the app into separate experiences.

Rank verification is human-reviewed and works identically across every game, including the three with no developer API (MLBB, CoDM, and Valorant's gated key). A player fills a per-game form, then attaches an in-game screenshot as proof. The proof lands in Supabase Storage under RLS; the submission enters a moderation queue where an admin approves or rejects it with a reason. Approval flips the profile badge to **Verified** and seeds the team's starting rating. From that point forward, dual-confirmed match results — not anyone's API — move the rating via an in-database Elo trigger. A trusted-submitter fast-track auto-approves players whose recent submissions were all clean, keeping the review queue small as the platform grows.

Scope discipline. The team has a 48-hour onsite build in Stage 2, during which a surprise theme or API is revealed and must be integrated. The architecture is deliberately modular — feature folders, a schema that tolerates new entity types, no clever infrastructure only one person understands — so an unknown requirement can be absorbed without a rewrite.

Tech Stack

Locked by the team — chosen for a one-service backend and a 48-hour build window.

LAYER	CHOICE	WHY
Frontend	Next.js 16 (App Router) · React 19 · TypeScript · Tailwind v4	One codebase; PWA now, native app later. Turbopack build.
Data	Supabase Postgres + Row-Level Security	Auth, storage, and realtime in one managed service — no separate backend to maintain.
Auth	Supabase Auth + roles[] claim	@*.edu.ph domain check establishes the collegiate tie; manual COR review as fallback.
Storage	Supabase Storage	Rank-proof screenshots; RLS limits reads to the owner plus anything approved and public.
Realtime	Supabase Realtime	Scrim notifications, announcements, chat.
Rating engine	In-DB Elo trigger on confirmed matches	No external API; identical logic across all four games.
Push	Web Push API	Scrim requests and match reminders, delivered through the installed PWA.
Hosting	Vercel	Preview deploys wired to the repo from hour one; GitHub-connected Supabase migrations auto-apply on push.

Account roles

Sign-up asks the user to pick a role, and the choice shapes the interface from that point on. Implementation is one `User` row with a `roles` array — not three separate account types — and role is switchable later from settings.

Player
Game profiles and ranks, joins teams, appears on player leaderboards, browses scrim, posts to the free-agent board.

Team Handler
Creates and manages a team, posts scrim listings and availability, sends/accepts scrim requests, reports results.

Both
A player-captain — the common case in the collegiate scene, and expected to be the most-selected option.

Core data entities

User (roles[])

School

Game (rank-tier config)

Team

GameProfile

RankSubmission

ScrimListing

ScrimMatch

Rating

Event

Announcement

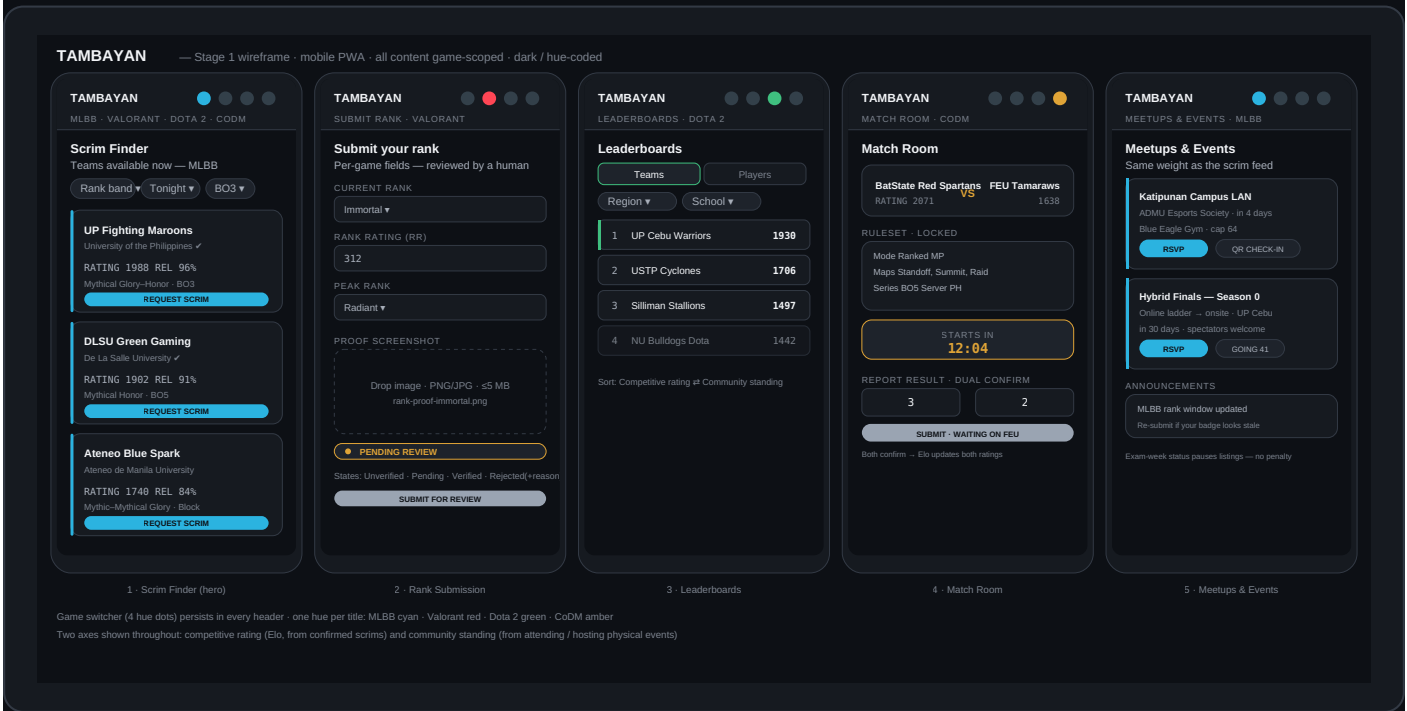
MentorshipRequest

FreeAgentPost

ReliabilityScore

UI Wireframe

Mobile-first — the product ships as an installable PWA. Dark, dense, high-contrast: the visual language competitive players already expect, not a generic social-app look.



Screen notes

- **1 · Scrim Finder (hero)** — filterable feed of available teams in the selected game: rank band, time window, format. Each card shows team name, school, rating, reliability score, verified badge, and a one-tap Request Scrim.
- **2 · Rank Submission** — per-game form plus screenshot upload, with pending / verified / rejected status states. The trust mechanism the whole rating system rests on.
- **3 · Leaderboards** — team ⇌ player toggle within the game, filterable by region and school, sortable by competitive rating or community standing.
- **4 · Match Room** — the booked scrim: agreed ruleset locked, opponent contact, live countdown, and the dual-confirmation result flow.
- **5 · Meetups & Events** — campus LANs, watch parties, and onsite finals with RSVP and QR check-in state, plus the game's announcement board — the same visual weight as the Scrim Finder, so the balance pillar is visible, not just claimed.